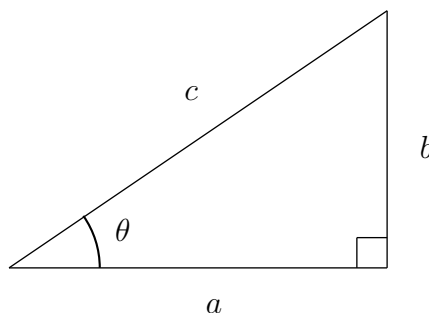

Notes for ‘Sine, cosine and tangent’

Important Ideas and Useful Facts:

- (i) The sine, cosine and tangent of an acute angle: Consider a right-angled triangle with acute angle θ , adjacent side length a , opposite side length b and hypotenuse c .



Define the *sine* of the angle θ to be

$$\sin \theta = \frac{b}{c} = \frac{\text{opposite}}{\text{hypotenuse}},$$

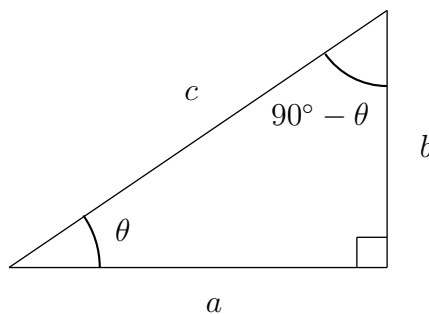
the *cosine* of θ to be

$$\cos \theta = \frac{a}{c} = \frac{\text{adjacent}}{\text{hypotenuse}},$$

and the *tangent* of θ to be

$$\tan \theta = \frac{b}{a} = \frac{\text{opposite}}{\text{adjacent}} = \frac{b/c}{a/c} = \frac{\sin \theta}{\cos \theta}.$$

Observe that the remaining angle of the triangle is $90^\circ - \theta$ (since the angles of the triangle add up to 180°).



It follows from the definitions that

$$\sin(90^\circ - \theta) = \frac{a}{c} = \cos \theta, \quad \cos(90^\circ - \theta) = \frac{b}{c} = \sin \theta,$$

and

$$\tan(90^\circ - \theta) = \frac{a}{b} = \frac{a/c}{b/c} = \frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta}.$$

For example,

$$\sin(70^\circ) = \cos(20^\circ), \quad \sin(20^\circ) = \cos(70^\circ), \quad \text{and} \quad \tan(70^\circ) = \frac{1}{\tan(20^\circ)}.$$

- (ii) **Notation for powers of sine, cosine and tangent:** To avoid brackets and to decongest formulae and expressions, one normally writes

$$\sin^n \theta = (\sin \theta)^n, \quad \cos^n \theta = (\cos \theta)^n \quad \text{and} \quad \tan^n \theta = (\tan \theta)^n,$$

for integers n greater than or equal to 2. In particular,

$$\sin^2 \theta = (\sin \theta)^2, \quad \cos^2 \theta = (\cos \theta)^2 \quad \text{and} \quad \tan^2 \theta = (\tan \theta)^2.$$

- (iii) **The circular identity:** For any angle θ , the following equation holds:

$$\sin^2 \theta + \cos^2 \theta = 1,$$

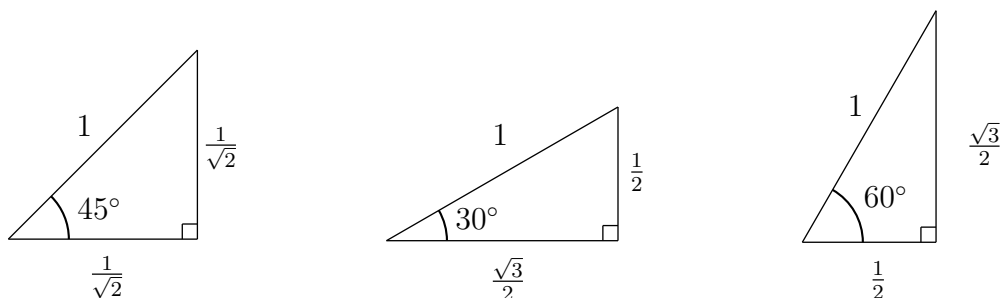
known as the *circular identity* (because of a close relationship to circles, explained later). This follows by the Theorem of Pythagoras applied to the definitions and the right-angled triangle above, noting that

$$b^2 + a^2 = c^2,$$

so that

$$\sin^2 \theta + \cos^2 \theta = \left(\frac{b}{c}\right)^2 + \left(\frac{a}{c}\right)^2 = \frac{b^2 + a^2}{c^2} = \frac{c^2}{c^2} = 1.$$

- (iv) **Values for common angles 45° , 30° and 60° :** The three simplest right-angled triangles use acute angles 45° , 30° and 60° respectively, drawn here with hypotenuse of length 1:



These produce the following values:

$$\sin 45^\circ = \frac{1}{\sqrt{2}}, \quad \cos 45^\circ = \frac{1}{\sqrt{2}}, \quad \tan 45^\circ = 1,$$

$$\sin 30^\circ = \frac{1}{2}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}, \quad \tan 30^\circ = \frac{1}{\sqrt{3}},$$

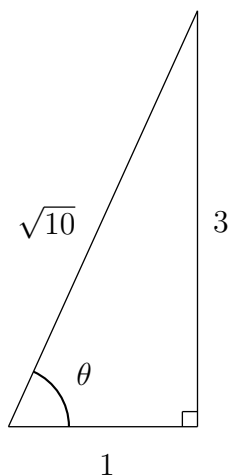
$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \quad \cos 60^\circ = \frac{1}{2}, \quad \tan 60^\circ = \sqrt{3}.$$

Examples:

1. Suppose that θ is an acute angle such that $\tan \theta = 3$. Find $\sin \theta$ and $\cos \theta$.

Solution: A diagram (even if just rough and ready and not necessarily accurate) and use of the Theorem of Pythagoras help us solve this problem quickly. We don't even need to find the angle θ .

We draw a right-angled triangle and label the sides opposite and adjacent to the angle θ with two numbers whose ratio is 3, which becomes the given $\tan \theta$. Because we can rescale any triangle, we can choose the adjacent side to have length labelled by 1, so the opposite side should have length labelled by 3.



By Pythagoras, the hypotenuse must have length labelled by

$$\sqrt{1^2 + 3^2} = \sqrt{1 + 9} = \sqrt{10} ,$$

added to the diagram above. We can now read off the following answers:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{3}{\sqrt{10}} ,$$

and

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{1}{\sqrt{10}} ,$$

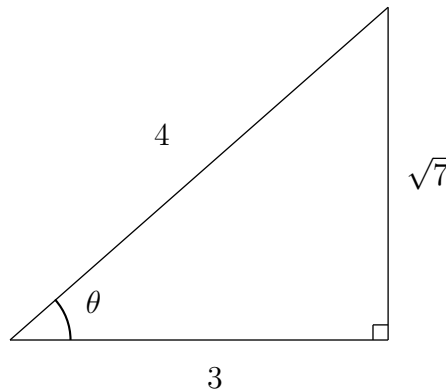
Just as a final check against errors, these expressions should satisfy the circular identity, which they do:

$$\sin^2 \theta + \cos^2 \theta = \left(\frac{3}{\sqrt{10}} \right)^2 + \left(\frac{1}{\sqrt{10}} \right)^2 = \frac{9}{10} + \frac{1}{10} = \frac{10}{10} = 1 .$$

2. Suppose that θ is an acute angle such that $\cos \theta = \frac{3}{4}$. Find $\sin \theta$ and $\tan \theta$.

Solution: Again, a diagram (even if just rough and ready and not necessarily accurate) and use of the Theorem of Pythagoras help us solve this problem quickly. Again, we don't need to find the angle θ .

We draw a right-angled triangle and label the side adjacent to the angle θ and the hypotenuse with two numbers such that their ratio is $\frac{3}{4}$, which becomes the given $\cos \theta$. Because we can rescale any triangle, we can choose the adjacent side to have length 3, and the hypotenuse to have length 4.



By Pythagoras, the opposite side must have length

$$\sqrt{4^2 - 3^2} = \sqrt{16 - 9} = \sqrt{7},$$

which has been added to the diagram above. We can now read off the following answers:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{\sqrt{7}}{4},$$

and

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sqrt{7}}{3}.$$